

MUHAMMAD ALZAGHLOUL

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience developing and evaluating AI solutions across Computer Vision, Machine Learning, Deep Learning, and NLP. Project experience spans instance segmentation, object detection, image classification, regression, and statistical data analysis. Skilled in model training and fine-tuning, dataset curation, experimental design, ablation studies, quantitative benchmarking, and error analysis, with a strong focus on research-driven development and rigorous model evaluation.

EDUCATION

Damascus University - Faculty of Informatics Technology

B.Sc. in Artificial Intelligence (2026)

TECHNICAL SKILLS

- **Languages:** Python, Dart
- **ML / DL & CV Stack:** PyTorch, Scikit-learn, OpenCV, YOLO / YOLO-Seg, SAM, MediaPipe, Hugging Face Transformers
- **Models & Methods:** CNN Architectures, ViT-based Models, Transfer Learning, Fine-Tuning, Feature Fusion, Ensemble Methods, Boundary Loss, Class-Imbalance Handling, ResNet50, AraBERTv2, Linear Regression
- **Data & Experimentation:** NumPy, Pandas, SciPy, Matplotlib, Seaborn, Dataset Curation, Feature Engineering, Cross-Validation, Ablation Studies, Model Evaluation, Error Analysis, Statistical Analysis
- **Backend & Mobile:** FastAPI, REST APIs, Flutter, BLoC, Dio
- **Tools & Platforms:** Git/GitHub, Jupyter Notebook, Google Colab, Kaggle, VS Code
- **Core AI Areas:** Computer Vision: Object Detection, Instance Segmentation, Image Classification, Image Processing, Machine Learning: Regression, Classification, Deep Learning: CNNs, ViTs, NLP: Transformers

KEY PROJECTS

Vehicle Inspection System (VIS) - Multi-View Vehicle Inspection | Graduation Project, 2026

YOLO26L-Seg • SAM • PyTorch • OpenCV • Instance Segmentation • Object Detection • Ensemble Voting

- Built an **end-to-end multi-view vehicle inspection system** combining **20-class vehicle-part instance segmentation** with **8-class external-damage detection and segmentation**.
- Extended a **SAV-based segmentation methodology** with **SAM-assisted mask refinement**, confidence-aware selection, and hybrid refinement for challenging vehicle regions.
- Benchmarked the proposed approach against **Mask R-CNN, GCNet, PANet, CBNet, and HTC**, achieving **68.5 AP50-95, 90.9 AP50, and 73.4 AP75 - 13.1 AP points above the strongest evaluated baseline**.
- Developed the damage pipeline using specialized YOLO detectors, instance segmentation, multi-scale processing, **Boundary Loss**, and ensemble voting for small and irregular damage, achieving **72.0% F1 and 80.4% recall**.

Acne Severity Classification & Lesion Analysis | 2025

ResNet50 • MediaPipe • OpenCV • Adaptive-LoG • PyTorch • Transfer Learning

- Designed a **two-backbone ResNet50 architecture** that fuses features from original RGB images with an acne-enhanced **Adaptive-LoG representation**.
- Engineered a face-aware preprocessing pipeline using **MediaPipe Face Mesh, LAB/HSV analysis, CLAHE, adaptive thresholding, LoG filtering, and morphological processing** to emphasize acne-related skin regions.
- Fine-tuned ImageNet-pretrained backbones with **partial layer freezing and class-weighted training**, comparing **RGB-only, 4-channel, and dual-backbone variants**.
- Improved test accuracy from **82.6% to 87.38%**, achieving **0.8793 Macro F1**, with class-wise evaluation and confusion-matrix analysis.

Arabic Punctuation Restoration | 2025

AraBERTv2 • PyTorch • Hugging Face Transformers • Token Classification • NLP

- Built an **AraBERTv2 token-classification system** for Arabic punctuation restoration, implementing text normalization, Transformer tokenization, and **word-to-subword label alignment**.

- Addressed severe class imbalance using **weighted CrossEntropy loss and rare-label enrichment**, with checkpointing, per-class evaluation, and confusion-matrix analysis.
- Improved test **Macro F1 from 0.5036 to 0.7392**, achieving **0.9501 Weighted F1** on approximately **997K test tokens**, and evaluated generalization on external-domain Arabic data.

Car Price Prediction | 2025

Python • Scikit-learn • Linear Regression • Feature Engineering • Cross-Validation

- Prepared and cleaned the vehicle-price dataset using **EDA, IQR-based outlier clipping, categorical encoding, and train-only feature scaling**, then engineered predictive features including power-to-weight, average MPG, and power-to-size.
- Built a supervised **Linear Regression pipeline with Scikit-learn** and evaluated it using train-test validation and **5-fold cross-validation**, with coefficient and error analysis to assess model behavior and generalization.

Uganda RCT Data Analysis - A Mother's Voice | Coursework

Pandas • NumPy • SciPy • Matplotlib • Data Cleaning • EDA • Statistical Analysis

- Cleaned and linked multi-level **randomized-trial survey data** covering **5,512 families and 22,308 child records**, including missing-value normalization, structural-column removal, ID-based linkage, and treatment/district summaries.
- Conducted **exploratory and statistical analysis** using **Spearman correlation tests** to investigate relationships involving household decision-making, maternal education, income, and nutrition-related change variables.

Hashtag Social Media Application | 2025

Flutter • Dart • BLoC • Dio / REST APIs • Video Trimmer

- Developed an Arabic-first Flutter social application using **BLoC**, repository-based data access, **GetIt dependency injection**, and **GoRouter**, supporting authentication, profiles, search, follow interactions, and short-media workflows.
- Integrated **Dio REST APIs**, secure token storage, media upload workflows, and **on-device video trimming**, together with like, save, report, and profile-content features.

LANGUAGES

- **Arabic.**
- **English.**